UNIVERSITY OF CAPE TOWN

Department of Electrical Engineering

EEE3096S

Practical 1B Report

Embedded Systems Timing, Optimisation and Interfacing

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1 Golden Measure and PC Timing

1.1 Objective

The objective of Task 1 was to establish a reference implementation of the integer square-root function and verify its correctness using a set of predefined golden input/output pairs. The execution time of the function was then measured on the host computer.

1.2 Golden Output Verification

$$\text{isqrt}(x) = \left\lfloor \sqrt{x} \right\rfloor$$

Table 1: Golden input/output verification

Input x	$\lfloor\sqrt{x}\rfloor$	**Verification**
0	0	$0^2 \leq 0 < 1^2$
1	1	$1^2 \leq 1 < 2^2$
15	3	$3^2 \leq 15 < 4^2$
16	4	$4^2 \leq 16 < 5^2$
4095	63	$63^2 \leq 4095 < 64^2$
65535	255	$255^2 \leq 65535 < 256^2$
123456789	11111	$11111^2 \leq 123456789 < 11112^2$
987654321	31426	$31426^2 \leq 987654321 < 31427^2$
4294836225	65535	$65535^2 \leq 4294836225 < 65536^2$
4294967295	65535	$65535^2 \leq 4294967295 < 65536^2$

All ten golden outputs passed verification.

1.3 Hand Verification

For $x = 987654321$:

$$31426^2 = 987,592,276$$
$$31427^2 = 987,655,129$$

$$987,592,276 \leq 987,654,321 < 987,655,129$$

$$\boxed{\left\lfloor \sqrt{987654321} \right\rfloor = 31426}$$

1.4 Timing Runs

Table 2: Host timing measurements

Run	**Repetitions**	**Time per Call (ns)**
Run 1	1,000	3.656
Run 2	1,000,000	1.833

The measured time per call decreases as the number of repetitions increases, indicating that the overhead of the timing mechanism becomes negligible for larger repetition counts.

1.5 Why a Single Function Call Is Unreliable

Measuring a single function call is unreliable because the execution time can be comparable to the resolution and overhead of the timing mechanism itself. Operating-system scheduling, cache state, branch prediction, and other processor activity can introduce variation. Repeating the function many times increases the measured interval and reduces the relative effect of these sources of timing uncertainty.

Task 1 Summary

- Golden output for 987654321: **31426**
- All golden tests: **PASS**
- Measured time per call: **1.833 ns** (1,000,000 repetitions)

2 Board Implementation and Timer Measurement

2.1 Objective

The objective of Task 2 was to transfer the integer square-root implementation to the STM32F051C8 development board and measure its execution time using the TIM16 hardware timer.

2.2 Required Enable Registers

- RCC->AHBENR: GPIO Clock enable register.
- RCC->APB2ENR: TIM16 clock enable register.

2.3 Clock Path Configuration

Table 3: System clock and TIM16 configuration

Clock Stage	Divider / Setting	Frequency
HSI source	–	8 MHz
AHB prescaler	/1	8 MHz
APB1 prescaler	/1	8 MHz
TIM16 prescaler	PSC = 7	1 MHz

$$f_{\text{timer}} = \frac{8\text{MHz}}{7+1} = 1\text{MHz}$$

$$\boxed{1 \text{ timer tick} = 1\mu\text{s}}$$

2.4 Predicted Execution Time

The integer square-root function uses a binary-search algorithm with a maximum of approximately 16 iterations. Based on an estimated 10–15 instructions per iteration, the predicted execution time was approximately:

$$\boxed{176 \text{ ticks} = 176\mu\text{s}}$$

2.5 Single-Call Timer Measurement

The timer count was sampled immediately before and after a single `isqrt()` call. The measured values are shown in Table 4.

Table 4: Single-call TIM16 measurement

Measurement	**Value**
Start count	35,822
End count	35,990
Count span	168 ticks
Execution time	168 μs

Since the timer frequency is 1 MHz:

$$t = \frac{168}{1 \times 10^6} = \boxed{168\mu\text{s}}$$

2.6 Oscilloscope Measurement

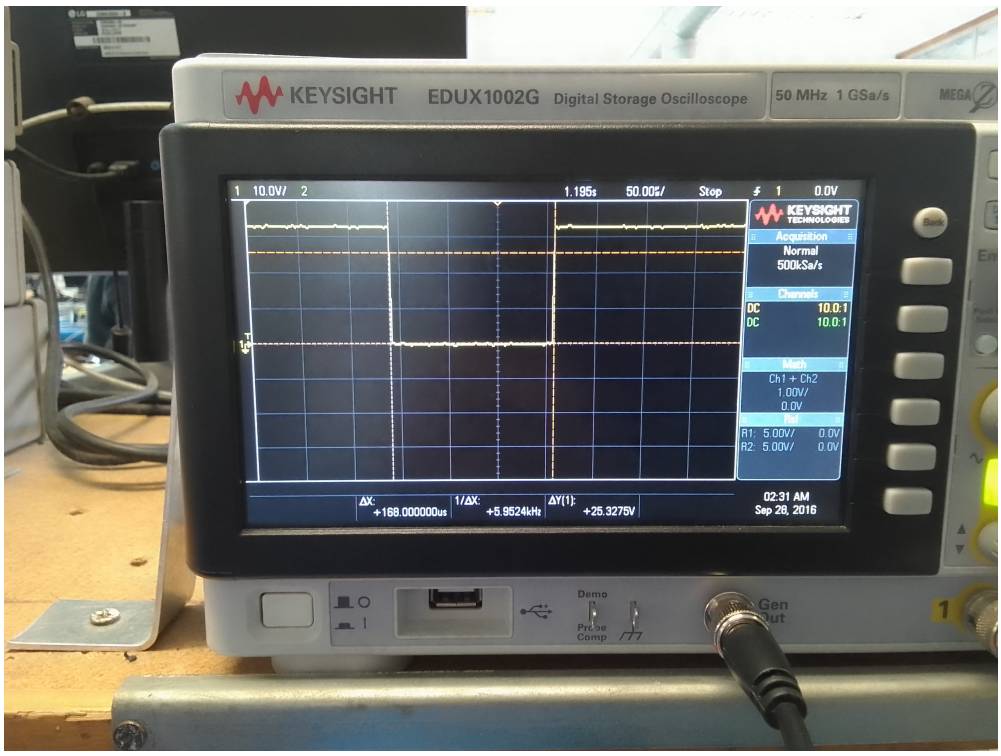


Figure 1: Oscilloscope measurement of the PC13 timing pulse for a single `isqrt()` call. The measured pulse width is approximately 168 μs.

Figure 1 shows the timing pulse generated on PC13. The measured pulse width is approximately 168 μs, which agrees with the timer-derived measurement.

2.7 Timer Wrap-Around

For the long-run measurement, 20,000 calls were executed while TIM16 was allowed to overflow. The timer is a 16-bit counter, so its maximum count is 65,535 and one complete timer period contains 65,536 ticks.

Table 5: Long-run timing measurement

Parameter	Value
Number of calls	20,000
Timer overflows	51
Final partial span	20,000 ticks
Total elapsed time	3,362,336 ticks
Mean time per call	168.117 μs

The total elapsed count is calculated using the number of complete timer overflows together with the final partial interval:

$$N_{\text{total}} = (51 \times 65536) + 20,000 = \boxed{3,362,336 \text{ ticks}}$$

Since one tick corresponds to 1 μs:

$$t_{\text{mean}} = \frac{3,362,336}{20,000} \times 1\mu\text{s} = \boxed{168.117\mu\text{s}}$$

2.8 Golden-Value Test

The ten predefined golden inputs were evaluated using the board implementation. All expected results were obtained, giving a **PASS** result. The pass/fail status is also indicated by the LED connected to PB1.

Task 2 Summary

- HSI frequency: **8 MHz**
- TIM16 prescaler: **7**
- Timer frequency: **1 MHz**
- Timer resolution: **1 μs**
- Single-call count span: **168 ticks**
- Single-call execution time: **168 μs**
- Long-run overflows: **51**
- Long-run mean execution time: **168.117 μs**
- Golden-value test: **PASS**

3 Compiler Optimisation

3.1 Objective

The objective of Task 3 was to investigate the effect of compiler optimisation on both execution time and program size.

3.2 Experimental Results

The measured results are presented in Table 6.

Table 6: Effect of compiler optimisation on execution time and code size

Flag	Text (B)	Data (B)	BSS (B)	Total (B)	Time (μs)	Ratio
-O0	4,644	12	1,580	6,236	358	1.00×
-O1	3,432	12	1,580	5,024	171	2.09×
-O2	3,460	12	1,580	5,052	180	1.99×
-Os	3,112	12	1,580	4,704	190	1.88×

The results show that increasing the optimisation level generally reduces both execution time and program size. The `-O1` build produced the shortest measured execution time at 171 μs, while `-Os` produced the smallest text section at 3,112 bytes.

3.3 Oscilloscope Measurements

Figures 2 through 5 show the oscilloscope measurements of the timing pulse for each optimisation level.

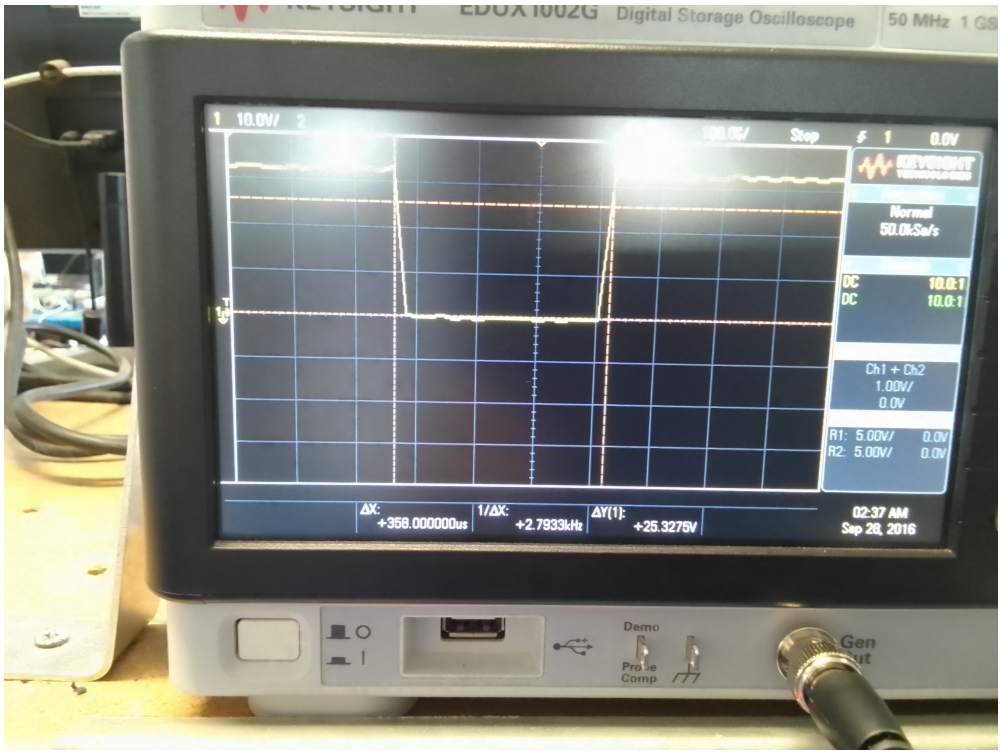


Figure 2: Oscilloscope measurement of the timing pulse with `-O0` optimisation. The measured pulse width is approximately 358 μs.

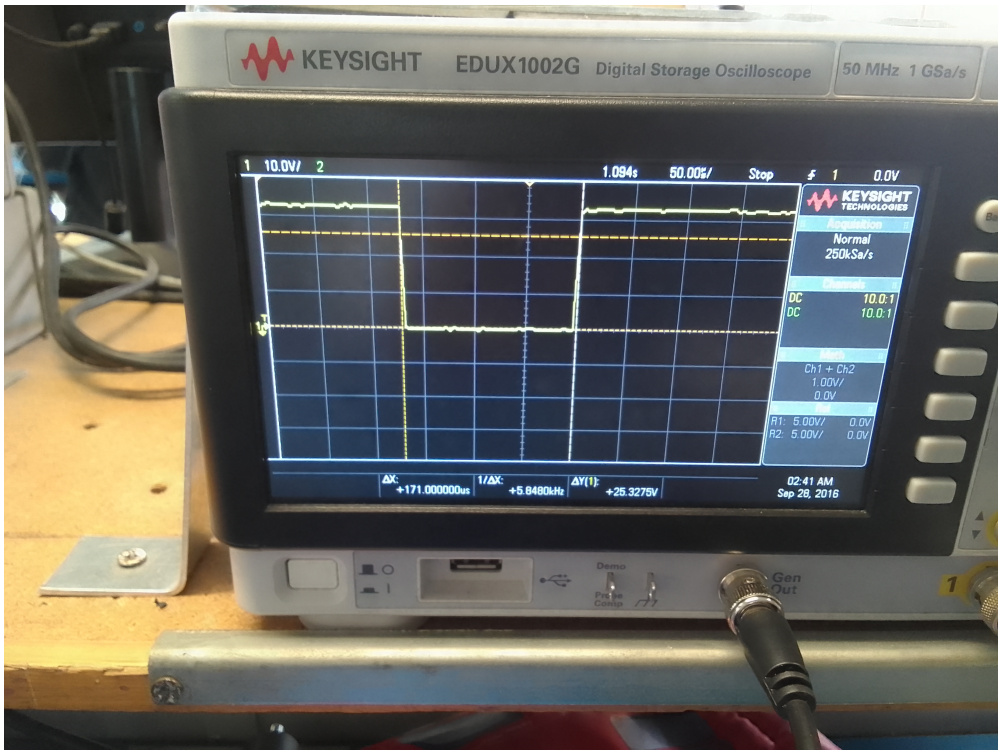


Figure 3: Oscilloscope measurement of the timing pulse with `-O1` optimisation. The measured pulse width is approximately 171 μs.

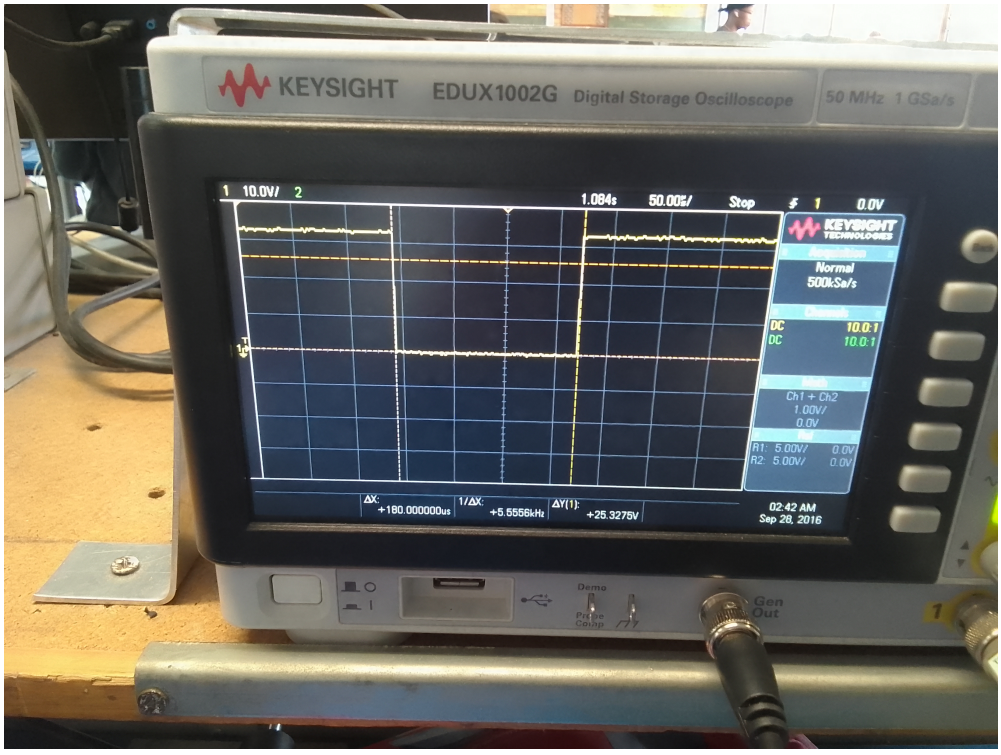


Figure 4: Oscilloscope measurement of the timing pulse with `-O2` optimisation. The measured pulse width is approximately 180 μs.

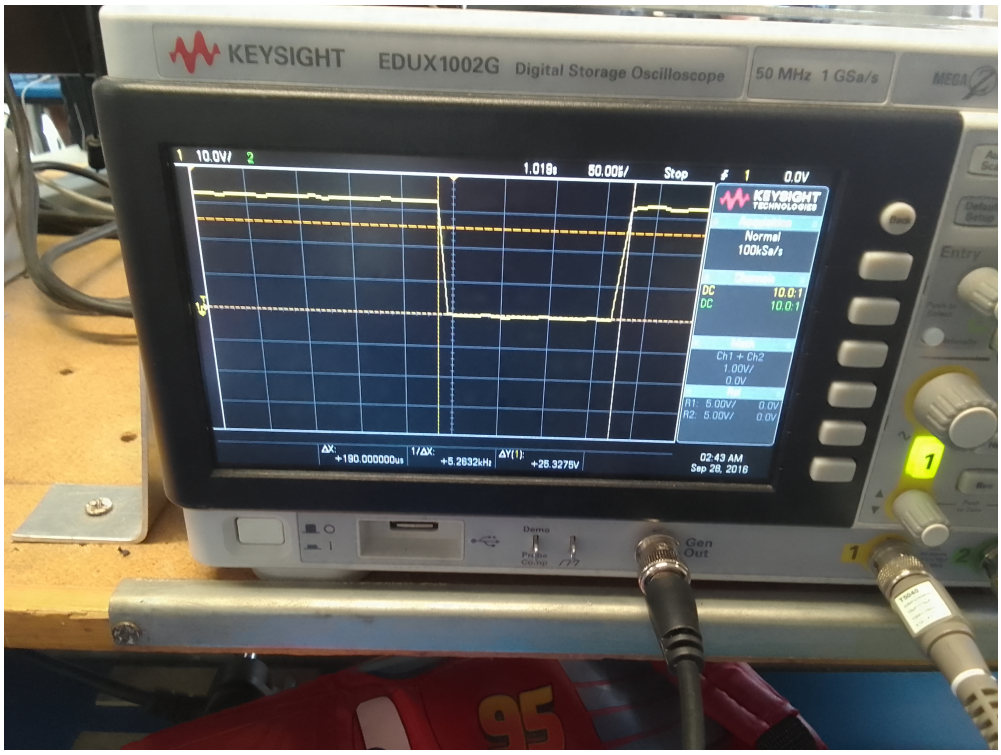


Figure 5: Oscilloscope measurement of the timing pulse with `-Os` (size optimisation) optimisation. The measured pulse width is approximately 190 μs.

3.4 Measured Speedup

$$\text{Speedup} = \frac{358}{171} \approx \boxed{2.09\times}$$

3.5 Optimisation Analysis

- **-O0:** Largest execution time at 358 μs and largest text section at 4,644 bytes. Useful for debugging.
- **-O1:** Significant improvement with execution time reduced to 171 μs (52% reduction) and text size reduced to 3,432 bytes (26% reduction).
- **-O2:** Execution time of 180 μs, slightly slower than -O1, with text size of 3,460 bytes.
- **-Os:** Smallest text section at 3,112 bytes (33% reduction from -O0) with execution time of 190 μs.

3.6 Compiler Transformation

From the disassembly, it is observed that no symbols for the ‘isqrt‘ function exist. From inspecting the call sites, it is evident that the function has been completely inlined by the optimizer.

Task 3 Summary

- Measured time at `-O0`: **358** μ**s**
- Measured time at `-O1`: **171** μ**s** (fastest)
- Measured speedup: **2.09**×
- Smallest text section: `-Os` at **3,112 bytes**

4 Cycle-Counted Phase Delay

4.1 Objective

The objective of Task 4 was to implement a cycle-counted delay in assembly so that the DAC output is delayed relative to a 1 kHz input signal by 45 degrees.

4.2 Required Phase Delay

$$f_{\text{input}} = 1\text{kHz}, \quad T_{\text{input}} = 1\text{ms}$$

$$\Delta t = \frac{45^\circ}{360^\circ} \times 1\text{ms} = \frac{1}{8} \times 1\text{ms} = \boxed{125\mu\text{s}}$$

At 8 MHz: $T_{\text{cycle}} = 125$ ns

$$N_{\text{cycles}} = \frac{125\mu\text{s}}{125\text{ns}} = \boxed{1000 \text{ cycles}}$$

4.3 Cycle Budget

The delay loop consists of a counter initialised to 249, with each iteration executing a decrement, two NOP instructions, and a conditional branch. The branch is taken for the first 248 iterations and not taken on the final iteration.

Table 7: Cycle budget for the delay loop

Instruction	**Cycles**	**Count**	**Subtotal**	**Running Total**
LDR (ADC read)	2	1	2	2
STR (DAC write)	2	1	2	4
MOVS	1	1	1	5
Delay Loop				
SUBS (first 248)	1	248	248	253
BNE (taken, first 248)	3	247	247	501
NOPS	3	3	3	3
BNE (not taken, last)	2	1	2	2
B loop	2	1	2	2

The remaining cycles required to obtain the complete 1000-cycle budget were implemented using additional NOP instructions.

4.4 Oscilloscope Measurement

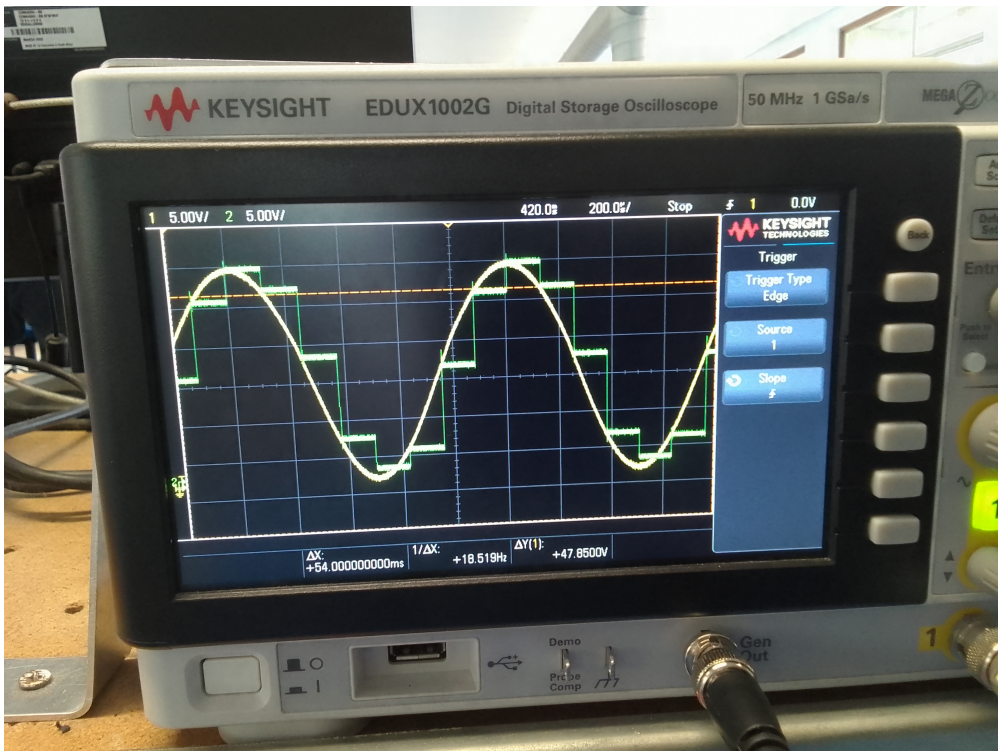


Figure 6: Oscilloscope measurement of the input and delayed DAC output.

Table 8: Measured phase-delay results

Parameter	Value
Input frequency	1.000 kHz
Output frequency	1.000 kHz
Expected delay	125.000 μs
Measured delay	125.000 μs

Task 4 Summary

- Input frequency: **1 kHz**
- Required phase shift: **45 degrees**
- Required delay: **125 μs**
- Required cycle budget: **1000 cycles**
- Measured delay: **125.000 μs**

5 Analog Debugging of the LCD Enable Line

5.1 Objective

The objective of Task 5 was to investigate the behaviour of the LCD enable signal after level shifting, identify the cause of an insufficient logic-high level, and modify the timing of the enable pulse.

5.2 Failing Case

The PC15 signal on the 3.3 V MCU side transitions correctly, but the level-shifted PC15_S signal does not reach the required 3.5 V logic-high threshold within the available pulse duration. The measured maximum voltage was approximately 2.5 V, as shown in Figure 7.

5.3 Oscilloscope Measurement

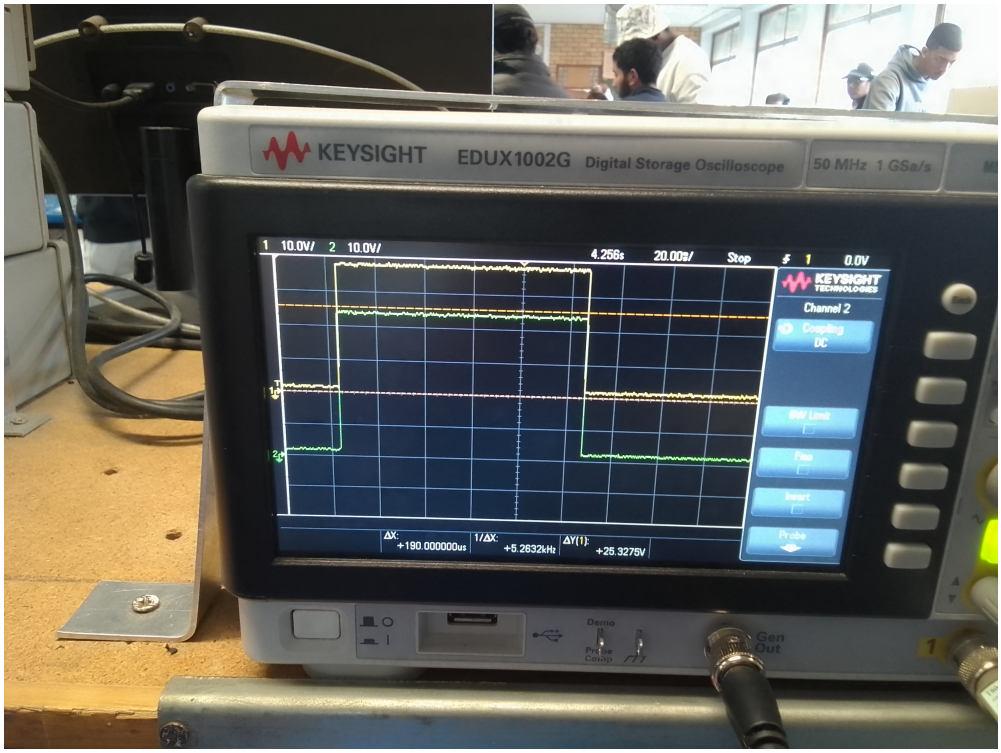


Figure 7: Oscilloscope measurement showing the PC15_S signal rise time. The measured pulse width is approximately 190 μs.

The rise time from the low level to the 3.5 V logic threshold was measured as approximately 2.0 μs.

5.4 Capacitance Estimate

The level-shifting network was approximated as a first-order RC charging circuit.

$$R = 1\text{k}\Omega, \quad V_{\text{CC}} = 5.0\text{V}, \quad V_C(t) = 3.5\text{V}, \quad t = 2.0\mu\text{s}$$

For an RC charging circuit:

$$V_C(t) = V_{\text{CC}} \left(1 - e^{-t/(RC)}\right)$$

Rearranging for capacitance:

$$C = \frac{-t}{R \ln\left(1 - \frac{V_C(t)}{V_{\text{CC}}}\right)}$$

Substituting the measured values:

$$C = \frac{-2.0 \times 10^{-6}}{1000 \ln(0.3)} \approx \boxed{1.66\text{nF}}$$

5.5 NOP Count Calculation

The measured rise time was approximately 2.0 μs. The required enable high-time contribution was 0.45 μs.

$$t_{\text{required}} = 2.0\mu\text{s} + 0.45\mu\text{s} = 2.45\mu\text{s}$$

At an 8 MHz clock, one processor cycle corresponds to 125 ns:

$$N = \frac{2.45\mu\text{s}}{125\text{ns}} = 19.6 \approx \boxed{20 \text{ NOPs}}$$

Twenty NOP instructions were therefore inserted between asserting and de-asserting the enable signal.

5.6 Fixed Case

After adding the additional delay, the level-shifted PC15_S signal reaches approximately 5.0 V and remains above the required 3.5 V threshold for the necessary duration.

5.7 LCD Verification

The final LCD output after the timing correction successfully displays the character "A".

Task 5 Summary

- Initial PC15_S maximum: **2.5 V** (required: 3.5 V)
- Measured rise time: **2.0 μs**
- Estimated capacitance: **1.66 nF**
- Timing correction: **20 NOPs**
- Final PC15_S high level: **5.0 V**
- LCD verification: **"A" displayed**

6 References

1. STMicroelectronics, "RM0091 Reference Manual: STM32F0x1/STM32F0x2/STM32F0x8 advanced ARM-based 32-bit MCUs," 2014.

2. STMicroelectronics, "STM32F051x4/x6/x8 Datasheet: ARM-based 32-bit MCU," 2013.

3. Hitachi, "HD44780U (LCD-II) Dot Matrix Liquid Crystal Display Controller/Driver Datasheet," 1998.